One policy per sequence, selected without ground truth

Track 2 -- Robotic Grounding. Luai Abuelsamen, University of California, Berkeley. Code: https://github.com/luaiabuelsamen/v2d-track2 (Apache-2.0). Container: see the form. Hardware: an NVIDIA Jetson for all CPU work; rented single-A6000 cloud boxes for training. *A longer version of this report, with the full derivations and the negative results, is `docs/TECHNICAL_REPORT_LONG.md` in the repository.*

Introduction

All three Track 2 tiers are scored against the same `multi_view_ground_truth`; only the permitted **input** differs. We found this by accident and then verified it: our Tier 1 and Tier 2 public scores came back identical, the two uploaded CSVs are byte-identical, and a Kabsch fit shows every shipped rollout's reference is an exact (0.000 cm) rigid transform of the released Tier 2 trajectory. Identical predictions earn identical scores.

That one fact set the strategy. The object trajectory is the deliverable; the policy exists to make it physically realisable; and the interesting engineering is not the reward but **what you are allowed to condition on**, because that is the only thing separating the tiers. We entered all three, and treated the input constraint as the design variable throughout: every per-clip decision in the Tier 2 and Tier 3 entries reads the *input* reference only, never the truth.

Approach

SOMA-X human parameters -> `soma_to_g1` retarget to a Unitree G1 with Dex3 hands (CPU, 30/30 clips at 0.0 mm residual against the released object trajectory) -> FlashCHORD SAC, **one policy per sequence**, 349,999,104 steps, `world_count=1`, `scene.support` set per clip from the shipped `support_surface` primitives -> `evaluate_policy.py` -> per-episode checkpoint selection -> the challenge kit's own packer.

Four recipe changes ship, each as a patch script that **refuses to apply to an upstream it does not recognise**, so a silent half-patch is impossible: an object-*orientation* reward term (weight 1.0, std 0.3) added to `recon_body`, which ships an `object_position_weight` and no orientation counterpart; a start-curriculum patch; object-position std tightened 0.2 -> 0.08 m on clips already tracking; and finger targets on bridge-smoothed references.

Selection is the submission rule, registered before the run. The provenance schema allows a different `checkpoint_sha256` per episode, so we pick, per clip, the latest checkpoint whose achieved object never leaves a 10 cm band around the *input* reference ("z-drop"). It reads only the input, so a reader with no ground truth can reproduce the pick. It cost us real score: 13 rollouts kit-score higher than the ones that ship, and we shipped the rule's picks anyway.

Tier 3 uses NVIDIA's own modules in their intended order -- `v2d_geocalib` -> `v2d_moge` -> `v2d_grounding_dino` -> a GroundingDINO-to-SAM2 prompt converter we had to write -> `v2d_sam2` -> `v2d_foundation_pose`. Placement in the released world is exactly one rigid transform taken from frame 0, as FAQ PR 207 permits: `world_from_cam = world_from_obj[0] @ inv(cam_from_obj[0])`, applied unchanged to every later frame, with no scale fitted and no per-frame correction. The frame-0 residual is 0.0000 cm and 0.000 deg on all 30 clips.

Warm-start fine-tuning (`brev/patch_warm_start.py`) adds `training.warm_start_actor_path`: after `create_learner` it loads a shipped actor's params and batch stats, sets `environment_steps` to 299,999,232 so the curriculum sits in its final stage, and runs to the normal 349,999,104 -- about 50M steps, a seventh of a full run, ~35 min and ~$0.57 on an A6000. The critic starts cold, so the actor's learning rate is scaled x0.1 for the first 2,000 updates.

Scoring was never reimplemented. `v2d score` hash-checks the kit's four metric files against `KIT_MANIFEST.json`, imports them and calls their `score()`. Across nineteen uploads the largest local-to-public disagreement in AUC was 1.1e-05, with the other three metrics matching exactly.

Experiments & Results

entry (30 clips)	AUC	SP-SR	MP-SR	MPPE cm	rank
Tier 1	**0.54808**	0.70000	0.50000	**12.060**	11 / 23
Tier 2	**0.46522**	0.66666	0.46666	**14.128**	6 / 18
Tier 3	**0.26781**	0.20000	0.03333	47.772	6 / 12
baseline, Tier 2	0.374	0.533	0.067	20.5	--
baseline, Tier 1	0.51038	--	--	--	--

Ranked by measured contribution rather than by interest:

`scene.support` per clip -- the largest single effect, and it was a bug. Our first full-length baseline scored mean AUC 0.043 because all three run scripts passed `scene.support=false` while NVIDIA's configs default it to `true`. Without the table the object falls at reset and the episode terminates on frame 0; metrics were bit-identical across all seven checkpoints, which is the signature. 14 of 30 clips ship a support primitive.

Warm-start fine-tuning is the one method change that paid, and only where the rules allow it. Tier 1 permits the clean multi-view trajectory, which the kit also names as the configured scoring reference, so a per-clip choice may legally read it. Fine-tuning shipped actors on that clean reference under a no-regression rule took Tier 1 from 0.46523 to **0.54808** AUC and MPPE from 14.128 to **12.060 cm**. On Tier 3 the input is our own reconstruction, so selection reads the input alone; 5 of 10 clips adopted that way, 0.23233 -> **0.25059**, turning both success gates off zero.

The mechanism is re-fitting to a new reference, not extra training. A cold swap fails outright: re-evaluating all 30 shipped actors on the clean Tier 1 trajectory *dropped* mean AUC from 0.46523 to 0.28937 and SP-SR from 0.667 to 0.138, with only 2 of 30 clips improving (frame checks <=1.7e-05 cm, so this is not a conversion artefact). A second generation of fine-tuning on the *same* reference adopted **0 of 3** and lost e16 outright (0.6469 -> 0.2963). Adoption is predicted by the clip's shipped AUC: 11 of 19 clips above 0.35 adopted at a mean gain of +0.169; **0 of 4** below 0.35 did.

Checkpoint selection is worth less than it looks, and an earlier draft overstated it. Across all 30 clips with the kit scorer, z-drop is worth **+0.0010** (0.3908 vs 0.3898 final). An oracle that reads the truth -- the ceiling on *any* selection rule -- reaches only 0.4201. The rule earns its place by being reproducible without ground truth, not by the points it adds.

Where the remaining gap actually lives. Measured on trajectories rather than on AUC (which mixes position and orientation, and whose two defects cancel): the noisy Tier 2 reference is 1.94 cm from truth on average while our policies are 13.84 cm, and **19 of 30 clips end up worse than the reference they were given**. The position error is ours -- Tier 2 is policy-bound, not reference-bound. But only **9 of 30 clips ever leave a 10 cm band** around their reference. So nine clips drop catastrophically and the rest track a good reference a few centimetres loosely: **precision across many clips, not robustness on a few**, is where most of the deficit is. Tier 3 is the opposite -- eleven clips have *negative* headroom, the policy already tracking better than the reconstruction it was given.

Conclusion

A three-tier benchmark that varies only the permitted input rewards treating the input constraint as the design variable, and we think the transferable results are these: a per-episode selection rule that reads the input alone is both legal in every tier and reproducible by a reader with no ground truth; cold-swapping a cleaner reference into a converged policy *loses* score, while fine-tuning it for a seventh of a run recovers and exceeds it; and the residual failure is mid-episode release of a well-tracked object, which our own diagnosis traces to the actor being paid for contact it cannot observe -- `envs/recon_body.py` computes the contact state every step and then discards it, so the finger action through the frames before a drop is necessarily open-loop.

Economic viability. Total spend was **~$144 of cloud credit** for 30 policies across three tiers and nineteen uploads -- a full-length cell is ~$3.00 and a warm-start cell ~$0.57, so the marginal cost of a new sequence is about the price of a coffee. *Data flywheel.* The selection and adoption rules read only the input, so every new rollout can be judged and adopted without any annotation -- the loop closes without a human or a ground truth in it. *Complexity reduction.* One CLI, four patch scripts that refuse unknown upstreams, and a scorer that imports the kit's own metric code under a hash check, which is why local numbers match the board to five decimals. *Innovation.* The orientation reward term, the input-only z-drop rule, and warm-start transfer as a way to move one policy to a different reference cheaply.

Limitations, stated plainly: one seed; no ablation isolates the four recipe changes from one another, because the shipping run was chosen as a shipping run, knowingly, over an attribution run; and object inertia is a sim artefact we did not touch -- `soma_to_g1` writes every object URDF as 1 kg with inertia `diag(0.01)`, making the squeegee, cane and vacuum 6-14x too easy to spin, and those are among our worst clips.